



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTROTECHNOLOGY
NQF LEVEL 2

19 February 2025

This marking guideline consists of 6 pages.

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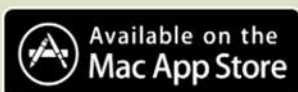
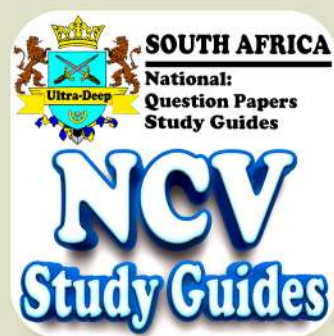
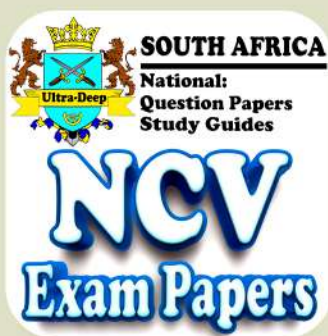


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QUESTION 1

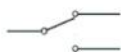
- 1.1 1.1.1 True
 1.1.2 False
 1.1.3 False
 1.1.4 False
 1.1.5 True
(5 × 1) (5)
- 1.2 1.2.1 $X_L = 2\pi fL$
 $= 2\pi \times 50 \times 0,2 \checkmark$
 $= 62,832 \Omega \checkmark$
- 1.2.2 $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 50 \times 10^{-6}} \checkmark = 63,662 \Omega \checkmark$
- 1.2.3 $Z = \sqrt{R^2 + (X_C - X_L)^2}$
 $= \sqrt{12^2 + (63,662 - 62,832)^2} \checkmark$
 $= 12,029 \Omega \checkmark$
- 1.2.4 $I_T = \frac{V}{Z} = \frac{230}{12,029} \checkmark = 19,12 A \checkmark$
(4 × 2) (8)
- 1.3 1.3.1 $\frac{1}{R_P} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4}$
 $\frac{1}{R_P} = \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$
 $\frac{1}{R_P} = \frac{11}{60} \checkmark$
 $\therefore R_P = 5,455 \Omega \checkmark$
 $R_T = R_1 + R_P + R_5$
 $R_T = 5 + 5,455 + 4 \checkmark$
 $R_T = 14,455 \Omega \checkmark$
(4)
- 1.3.2 $V_T = I_T \times R_T$
 $V_T = 12 \times 14,455 \checkmark$
 $V_T = 173,46 V \checkmark$
(2)
- 1.4 Joule's law states that the heat produced by the current flow in a conductor is directly proportional to the square of the current, the resistance and time.
(1)
[20]

QUESTION 2

2.1 2.1.1



2.1.2



2.1.3

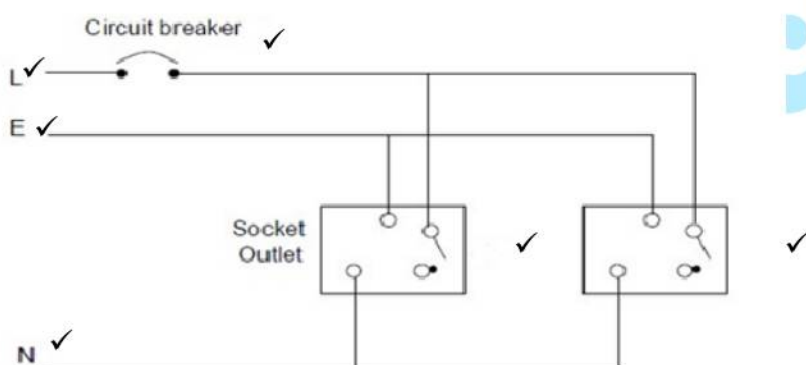


2.1.4



(4 × 1) (4)

2.2



(6)

2.3

2.3.1 4 mm²

2.3.2 6/8 mm²

2.3.3 10 mm²

2.3.4 2,5 mm²

2.3.5 1 mm²

(5 × 1) (5)

2.4

Advantages:

- Provides mechanical protection
- Lasts longer than PVC
- Can be extended easily

Disadvantages:

- Can corrode easily
- More expensive than PVC
- Time-consuming to install e.g. threading and bending

(6 × 1) (6)

2.5

2.5.1 A fault current is a current that results from an insulation failure or from bridging of insulation.

2.5.2 Earthing is the process by which all sub-circuits and metal parts in a building are linked to the ground.

(2 × 2) (4)

[25]

QUESTION 3

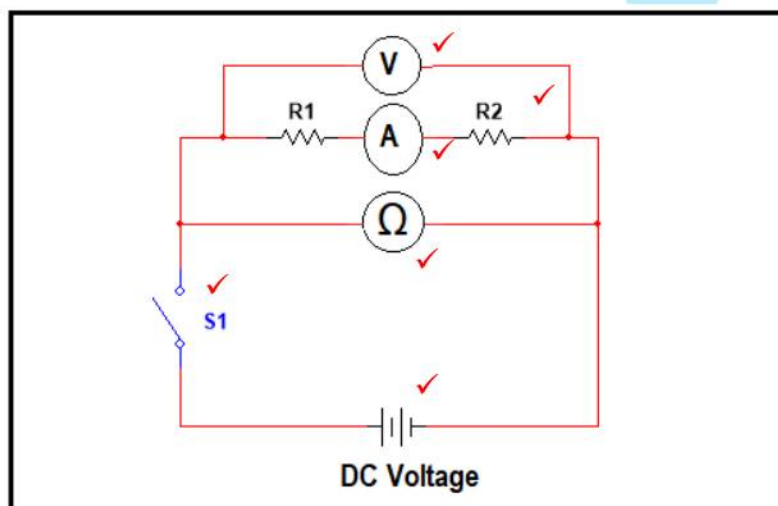
- 3.1 A – Centre-tapped transformer
B – Diode or PN junction diode
C – Polarised capacitor
D – Zener diode
E – Fixed resistor
F – NPN transistor

(6)

- 3.2
- Set the digital multi meter to diode test.
 - Connect each part of the transistor, N, P, and N to give a total of three connections.
 - With the two connections of a diode, only two resistance readings are taken, that is, in each direction across the two terminals.
 - With three terminals, six resistance readings are possible.

(4)

3.3



(6)

- 3.4
- What needs to be done?
 - Where does it have to be done?
 - How will it be done?
 - What resources – time, materials, and tools are available?

(4)

- 3.5
- Soldering iron and stand
 - A good or reliable multi meter
 - Screw drivers
 - Small side cutter
 - Small long-nose pliers
 - A utility knife/cable knife
 - Desoldering pump

(Any 5 × 1)

(5)
[25]

QUESTION 4

- 4.1 4.1.1 B
 4.1.2 E
 4.1.3 F
 4.1.4 A
 4.1.5 D

(5 × 1) (5)

- 4.2 4.2.1 High
 4.2.2 Low
 4.2.3 Unwanted
 4.2.4 Break or opening
 4.2.5 Continuity or connection

(5 × 1) (5)

- 4.3 4.3.1 $R_s = R_1 + R_2$
 $= 2 + 4$
 $= 6 \Omega \checkmark$
 $R_p = \frac{R_s \times R_3}{R_s + R_3}$
 $= \frac{6 \times 3}{6 + 3}$
 $= 2 \Omega \checkmark$
 $\therefore R_T = R_P = 2 \Omega \checkmark$
 $I_T = \frac{V_T}{R_T} = \frac{12}{2} = 6 A \checkmark$

(4)

- 4.3.2 $I_{A2} = \frac{V}{R_3} = \frac{12}{3} \checkmark = 4 A \checkmark$

(2)

- 4.3.3 $I_{A1} = \frac{V}{R_3} = \frac{12}{3} \checkmark = 4 A \checkmark$

(2)

- 4.3.4 • S1 could be in the open position
 • The ammeter is faulty (open circuit)
 • There is a possible break in the wire after ammeter A₁ (open circuit)
 • The battery could be totally discharged/flat/dead (Any 2 × 1)

(2)
[20]

QUESTION 5

5.1

TIME	WEEKS✓								
ACTIVITY	1	2	3	4	5	6	7	8	9
A			✓						
B							✓		
C						✓			
D								✓	
E ✓									✓

(1 mark for the chart)✓

(8)

5.2

Activity E will be completed on week 11.

(2)

[10]

TOTAL:

100